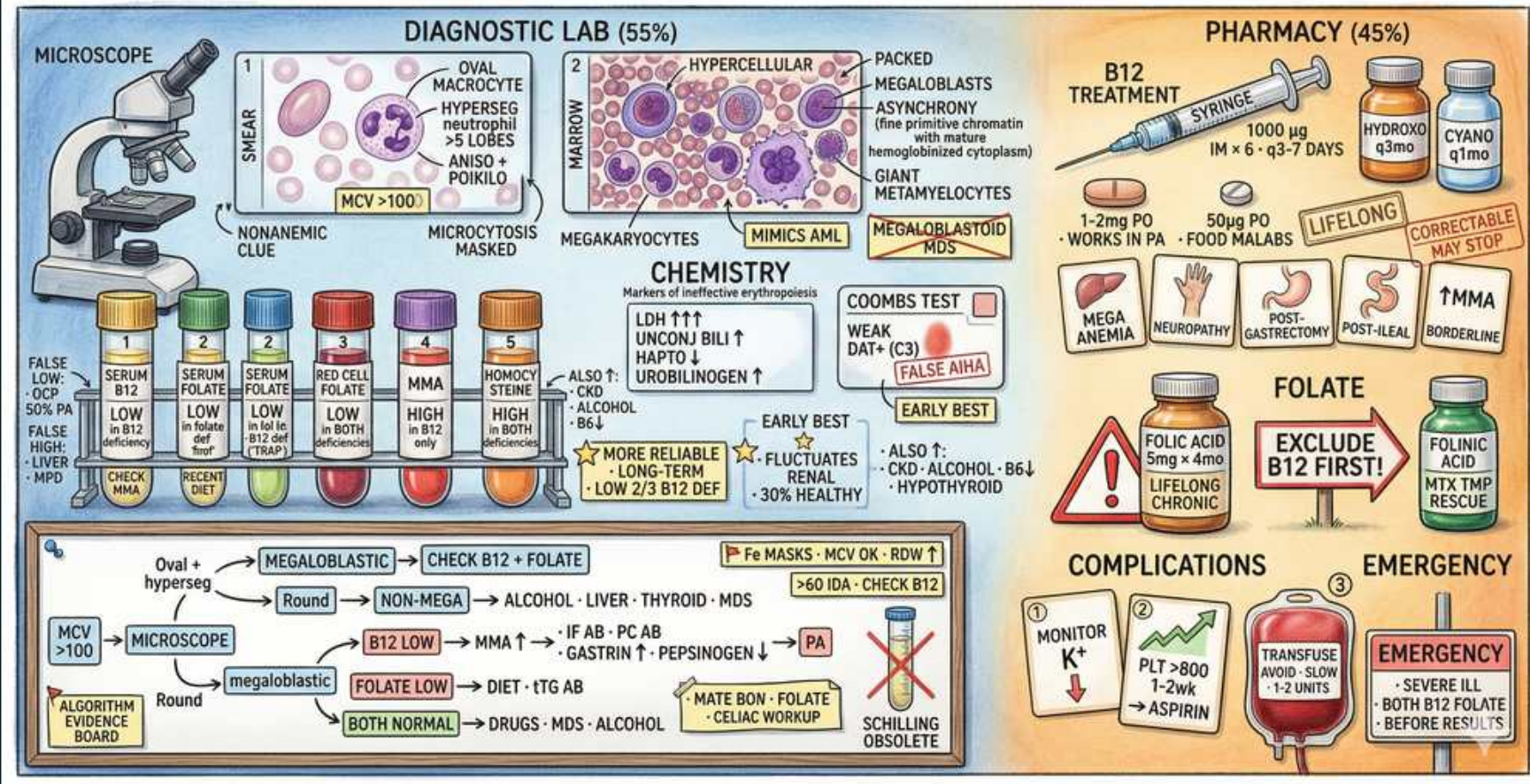


Megaloblastic — Workup & Pharmacy Overview

THE MEGALOBLASTIC WORKUP & PHARMACY



1. Smear: oval macrocytes

WHAT YOU SEE

MICROSCOPE smear 'OVAL MACROCYTE', 'HYPERSEG neutrophil >5 lobes', 'MCV >100'

WHAT IT ENCODES

Peripheral smear is the first cheap discriminator: OVAL macrocytes (macro-ovalocytes) and HYPERSEGMENTED neutrophils (≥5 lobes; ≥5% with 5 lobes or any with 6) define a MEGALOBLASTIC process, with MCV >100 fL (often >110-120). Round macrocytes/target cells without hypersegmentation suggest liver disease or alcohol instead. Anisopoikilocytosis and a low reticulocyte count accompany the maturation defect. Mnemonic: 'oval + hypersegmented = order MMA and homocysteine.'

2. Hypercellular marrow megaloblasts

WHAT YOU SEE

'HYPERCELLULAR', 'PACKED MEGALOBLASTS', 'GIANT METAMYELOCYTES', 'MIMICS AML'

WHAT IT ENCODES

Marrow (rarely needed) is HYPERcellular and packed with megaloblasts and pathognomonic GIANT METAMYELOCYTES/band forms; the open, immature chromatin can MIMIC acute leukemia (AML) or MDS — a classic pitfall, so don't biopsy-diagnose leukemia in an untreated B12/folate-deficient patient. Repeat after vitamin repletion if uncertain. Mnemonic: 'giant metamyelocytes = megaloblastic, NOT leukemia — replete first.'

3. Microcytosis masked

WHAT YOU SEE

'MICROCYTOSIS MASKED'

WHAT IT ENCODES

Coexisting IRON deficiency (or thalassemia/anemia of chronic disease) lowers MCV and can normalize it, MASKING the macrocytosis — a 'dimorphic' picture with a normal mean MCV but a wide RDW and two RBC populations on smear. Suspect this when an expected macrocytic anemia has a normal MCV plus high RDW; the smear and iron studies reveal it. Mnemonic: 'normal MCV + high RDW = two populations hiding each other.'

4. Serum B12 low

WHAT YOU SEE

CHEMISTRY tube 'SERUM B12 LOW'

WHAT IT ENCODES

Low serum B12 supports deficiency but is imperfect: it measures total (mostly inactive haptocorrin-bound) B12, so borderline-low values (~150-300 pg/mL) are unreliable and falsely low values occur in pregnancy, oral contraceptives, and folate deficiency. Confirm equivocal cases with MMA ± homocysteine. Mnemonic: 'serum B12 screens; MMA confirms.'

5. MMA high (more reliable)

WHAT YOU SEE

Tube 'MMA HIGH', 'MORE RELIABLE'

WHAT IT ENCODES

Methylmalonic acid (MMA) is the most reliable functional marker of B12 deficiency: it rises EARLY and is the metabolite of the B12-dependent methylmalonyl-CoA mutase reaction. Crucially, MMA is HIGH in B12 deficiency but stays NORMAL in folate deficiency — this is the single best discriminator between the two. (MMA can also rise in renal insufficiency/volume depletion.) Mnemonic: 'MMA = B12's fingerprint; folate leaves no MMA mark.'

BOARD TRAP

Assuming MMA also rises in folate deficiency. It does NOT — MMA is specific to B12 and is precisely what separates B12 (high MMA + high homocysteine) from folate (normal MMA + high homocysteine). Picking 'high MMA' for isolated folate deficiency is the classic distractor.

6. Homocysteine high (both)

WHAT YOU SEE

Tube 'HOMOCYSTEINE HIGH'

WHAT IT ENCODES

Homocysteine rises in BOTH B12 and folate deficiency (the remethylation step needs both), so it is sensitive but NON-specific and cannot distinguish them alone. Combine with MMA: high MMA + high homocysteine = B12; normal MMA + high homocysteine = folate. Elevated homocysteine is also independently prothrombotic. Mnemonic: 'homocysteine = both vitamins; MMA breaks the tie.'

7. Ineffective erythropoiesis markers

WHAT YOU SEE

'LDH↑↑↑', 'UNCONJ BILI↑', 'HAPTO↓', 'URINOBILINOGEN↑'

WHAT IT ENCODES

Intramedullary destruction of precursors produces a 'hemolysis-like' panel without true peripheral hemolysis: markedly elevated LDH (often very high), raised UNCONJUGATED bilirubin, LOW haptoglobin, and increased urinary urobilinogen — yet the reticulocyte count is LOW (the marrow is ineffective). This is ineffective erythropoiesis, the hallmark of megaloblastic anemia. Mnemonic: 'looks like hemolysis (high LDH/bili, low hapto) but retics are LOW.'

8. Coombs negative

WHAT YOU SEE

'COOMBS TEST NEGATIVE (C3)', 'WEAK DAT'

WHAT IT ENCODES

The direct antiglobulin (Coombs) test is NEGATIVE in megaloblastic anemia — the high LDH/bilirubin and low haptoglobin come from intramedullary (non-immune) destruction, not antibody-mediated peripheral hemolysis. A negative DAT therefore distinguishes megaloblastic ineffective erythropoiesis from autoimmune hemolytic anemia (where DAT is positive). Mnemonic: 'hemolysis labs + NEGATIVE Coombs + LOW retics = think B12/folate, not AIHA.'

BOARD TRAP

Expecting a POSITIVE Coombs because the labs look hemolytic. The destruction here is intramedullary and non-immune, so Coombs/DAT is NEGATIVE — a positive DAT would instead point to autoimmune hemolytic anemia (which has a HIGH reticulocyte count, the opposite of megaloblastic anemia).

9. Non-megaloblastic causes

WHAT YOU SEE

ALGORITHM 'NON-MEGA → ALCOHOL, LIVER, THYROID, MDS'

WHAT IT ENCODES

After confirming macrocytosis (MCV >100), the smear splits the differential: NON-megaloblastic macrocytosis = alcohol, chronic liver disease, hypothyroidism, MDS, reticulocytosis (acute hemolysis/blood loss), and drugs — all WITHOUT hypersegmented neutrophils or high MMA. Workup: smear, reticulocyte count, LFTs, TSH, B12/folate, and consider MDS (cytopenias, dysplasia) if unexplained. Mnemonic 'FAT RBC': Fetus(pregnancy)/Folate, Alcohol/liver, Thyroid, Reticulocytosis, B12.

10. Schilling test obsolete

WHAT YOU SEE

'SCHILLING OBSOLETE'

WHAT IT ENCODES

The Schilling test (radiolabeled B12 ± intrinsic factor to localize the malabsorption) is OBSOLETE and no longer used — diagnose pernicious anemia instead with anti-intrinsic-factor (specific) and anti-parietal-cell (sensitive) antibodies, plus elevated gastrin/low pepsinogen I and endoscopy when needed. Picking Schilling as 'next step' is a dated distractor. Mnemonic: 'no more radioactive B12 — order anti-IF antibodies.'

11. B12 1000ug IM ×7 days

WHAT YOU SEE

PHARMACY 'B12 TREATMENT 1000ug IM ×6-7 DAYS', hydroxo/cyano

WHAT IT ENCODES

B12 replacement: 1000 µg IM (hydroxocobalamin preferred over cyanocobalamin) daily/alternate-day for ~1 week (loading), then maintenance — hydroxo ~q3 months, cyano ~q1 month. High-dose ORAL B12 (1000-2000 µg/day) is also effective even in PA via ~1% IF-independent passive absorption, useful for stable, adherent patients. IM is preferred for severe/neuro disease or unreliable absorption. Mnemonic: 'load IM 1000 µg, then maintain; oral works in PA too.'

12. Exclude B12 before folate

WHAT YOU SEE

FOLATE warning 'EXCLUDE B12 FIRST!', folinic acid MTX rescue

WHAT IT ENCODES

Before giving folic acid for a macrocytic anemia you MUST exclude/treat B12 deficiency — folate alone corrects the anemia but lets B12 neuro disease (subacute combined degeneration) progress, potentially irreversibly. For methotrexate/trimethoprim/pyrimethamine toxicity (DHFR inhibitors) the rescue is FOLINIC acid (leucovorin), the already-reduced folate that bypasses the blocked enzyme — plain folic acid won't work there. Mnemonic: 'check B12 first; for DHFR blockers give folinic, not folic.'

BOARD TRAP

Giving folate alone for any macrocytic anemia 'because it's safe.' This is the trap: in unrecognized B12 deficiency, folate masks the anemia while neuro damage advances. Always exclude B12 first (or give both). Also wrong: using plain folic acid to rescue methotrexate toxicity — the DHFR block requires FOLINIC acid (leucovorin).

13. Monitor K+ on treatment

WHAT YOU SEE

COMPLICATIONS 'MONITOR K+'

WHAT IT ENCODES

During early repletion the burst of new erythropoiesis rapidly consumes potassium → HYPOKALEMIA (can be severe, with arrhythmia risk) in the first days, and may also unmask iron deficiency. Monitor and replace K+ (and iron) as the hemoglobin rises; a reticulocyte peak at ~1 week confirms response. Mnemonic: 'new cells eat potassium — watch the K+ during recovery.'

14. Emergency: severe anemia

WHAT YOU SEE

'EMERGENCY SEVERE ILL', transfuse 1-2 units slow

WHAT IT ENCODES

Severe symptomatic anemia (heart failure, hemodynamic compromise): draw B12/folate/MMA/homocysteine first, then start parenteral B12 + folate empirically WITHOUT awaiting results; transfuse only if truly necessary and then SLOWLY, 1-2 units with diuretic cover, because chronic megaloblastic anemia carries an expanded plasma volume (TACO/pulmonary edema risk). Vitamin repletion, not transfusion, is definitive. Mnemonic: 'sample, then treat now; transfuse slow if you must.'